

SQL Introduction – Revision Notes

Structured Query Language Fundamentals, History, Categories & Core Architecture

1. What is SQL?

SQL (Structured Query Language) is standard database domain language used to communicate with and perform operational commands on structured data in relational databases.

Fundamental Flow: User / Application → SQL Query → Database Engine (DBMS)

Core Operations

- Create database structures & tables
- Insert, update, and delete records
- Retrieve filtered and aggregated data

Control & Integrity

- Apply rules & operational constraints
- Control user access & permissions
- Manage transactional consistency

2. Understanding "Structured" Data

Data in a relational model is organized systematically into **tables (relations)** consisting of rows (tuples) and columns (attributes).

Student_ID (PK)	Name	City
101	Ravi	Delhi
102	Aman	Mumbai

3. History & Evolution of SQL ★

- **1970 – Relational Model:** Proposed by **E. F. Codd**. Data represented as mathematical relations (tables) using Relational Algebra & Tuple Relational Calculus.
- **IBM Implementation:** IBM developed an early relational database query language called **SEQUEL** (Structured English Query Language), later shortened to **SQL**.
- **Commercial Systems:** Oracle, Microsoft SQL Server, and IBM DB2 emerged as leading technologies based on SQL standards.

⚠ **Exam Landmark Chain:** E. F. Codd → Relational Model → IBM / SEQUEL → SQL

4. SQL as a Domain-Specific Language (DSL) ★

Unlike general-purpose programming languages designed for wide operational domains, SQL is specifically tailored for relational database management systems.

Language	Type	Primary Purpose

5. Declarative Paradigm & Golden Rule ★★☆☆

SQL is primarily a **Declarative Language**. You declare *WHAT* result you require, leaving the DBMS optimization engine to decide *HOW* to fetch it.

Declarative (SQL)

- Specifies **WHAT** data to retrieve.
- Focuses on final output requirements.
- No low-level algorithms specified by user.

Procedural (C / C++)

- Specifies **WHAT + HOW** step-by-step.
- Focuses on execution control flow.
- Requires explicit loops & file access details.

```
-- Declarative Query: User does not specify index scan or file reading steps
SELECT *
FROM Students
WHERE City = 'Delhi';
```

Golden Rule ★: SQL = WHAT | Procedural Language = WHAT + HOW

6. SQL vs. PL/SQL

PL/SQL (Procedural Language extension to SQL) is Oracle's proprietary extension adding control structures to standard SQL.

Feature	Standard SQL	PL/SQL
Paradigm	Declarative	Procedural Extension
Control Flow	Not supported natively	Supports IF-ELSE, Loops, Functions
Components	Queries, Statements	Procedures, Functions, Cursors, Triggers

7. Main SQL Command Categories

DDL

Definition

Defines structure.

CREATE, ALTER, DROP

DML

Manipulation

Modifies data.

INSERT, UPDATE,
DELETE

DCL

Control

Access privileges.

GRANT, REVOKE

TCL

Transaction

Manages units.

COMMIT, ROLLBACK,
SAVEPOINT

8. Keys & Constraints

Constraints enforce data integrity rules across relational tables.

```
CREATE TABLE Student (
  Student_ID INT PRIMARY KEY,
  Name VARCHAR(50) NOT NULL,
  City VARCHAR(50) DEFAULT 'Karnal';
```

9. Filtering Operators & Clause Order ★★☆☆

Essential query filters and logical clause precedence rules for execution.

```
-- Selective aggregation and filter example
SELECT City, COUNT(*)
FROM Student
WHERE Marks >= 50
GROUP BY City
HAVING COUNT(*) > 2
ORDER BY City;
```

Standard Clause Order: FROM → WHERE → GROUP BY → HAVING → ORDER BY

10. Aggregate Functions & Advanced Topics

Aggregate Functions

- MAX() / MIN(): High/low values
- AVG(): Arithmetic average
- COUNT(): Row tally
- SUM(): Cumulative total

High-Priority Exam Topics

- **Joins** ★★☆☆: Combine multi-table data
- **Subqueries** ★★☆☆: Nested SELECT queries
- **Correlated Queries**: Inner dependent queries

11. Real-World Application Example

IRCTC Ticketing System: Searching for train availability (Date → Train → Seats) executes backend APIs running optimized multi-table JOINS and nested SQL queries to return live status.

12. Syllabus Quick-Revision Mind Map

SQL Architecture & Fundamentals | — 1. Foundation Principles | — Domain-Specific (DS

 **One-Line Memory:** SQL = Structured + Relational + Domain-Specific + Declarative Query Language.